

Response to the reviewers

Aggressive feature reduction degrades the selective reliability and clinical net benefit of clinical risk models

Scientific Reports submission 2650209c-ff1e-4df4-aeb0-75ef3b35e564 (revised manuscript)

Greetings, and thank you — both reviews improved this paper materially, and Reviewer 1’s first point changed one of its conclusions. Rather than patch the affected analyses, we re-ran the entire experimental grid from scratch under a new, fully hash-pinned configuration, extended it with an exact- k budget arm and the full Diabetes-130 cohort, and re-derived every number in the manuscript from the regenerated results. The rebuild reproduces the originally submitted aggregates on the eleven unchanged datasets at machine precision for the tree-based learners ($\leq 1.2 \times 10^{-16}$ on any estimate or interval bound) and within 1×10^{-4} for logistic regression (largest observed deviation 5.1×10^{-5} ; a documented solver-version sensitivity, and the environment is now version-locked). The Holm-corrected p -values move correspondingly more on those logistic cells (at most 3.9×10^{-3}), and every one of the 5,265 materiality verdicts is nonetheless identical — so all changes below are attributable to the revision’s additions, not to instability of the pipeline.

Reviewer 1

1. The rounding rule retains at least one feature no matter how aggressive the cut... feature count and absolute retained- k are mechanically linked through this rule, so the correlation may substantially reflect the simpler point that “reducing to one feature is unsafe.” I ask the authors to re-relate the harm to the absolute number of retained features (or compare datasets at matched retained- k).

Response. You were right, and we did both — and the matched- k experiment overturned our original reading. First, the reanalysis you asked for: harm correlates with the absolute retained count *more strongly* than with the feature count (Spearman $\rho = -0.78$ vs. -0.71 ; both leave-one-dataset-out robust with bootstrap intervals excluding zero), and excluding the two floor-collapsed datasets attenuates the p -association to non-significance ($\rho = -0.59$, $p = 0.075$, $n = 10$). Second, the decisive experiment: we added an *exact- k budget arm* ($k \in \{1, 2, 4, 8\}$ for every dataset, evaluated inside the same repetitions and folds as the fractional budgets, in its own Holm family so the original comparisons are untouched). At matched $k = 1$, *every* dataset is materially harmed — including the wide ones: SPECTF ($p = 44$), whose own 25% budget costs +0.002 AURC, pays +0.035 at $k = 1$; WDBC +0.030; the full readmission cohort +0.027; Statlog and Cleveland pay +0.19 and +0.18. The claim you challenged (“datasets with more candidate features are reliably safer to reduce”) is therefore *retired* in the revision: the apparent robustness of wide datasets is a property of the retained count their fractional budgets never leave, dataset structure modulates severity at matched k by an order of magnitude, and the harm is not attributable to instability of the selection itself. On that last point we report a dissociation rather than lean on a correlation: Haberman is the second most stable dataset in the study and the third most harmed, while SPECTF is among the least stable and the least harmed. The correlational test is given in both directions and as measured — null across datasets ($\rho = +0.14$, $p = 0.66$), and at the finer (dataset, ranker) level null when pooled ($\rho = -0.047$, $n = 36$) but with a weak within-dataset trend ($\rho = -0.29$, $p = 0.19$ by block permutation) whose sign is the one a noise mechanism would predict. We state plainly that this trend is non-significant, that it is confounded with ranker quality, and that we do not rest the claim on it; the exact- k arm carries it instead, since at $k = 1$ even datasets with near-deterministic

selections are materially harmed. Section 4.7 is rewritten around this result, the budget rule’s floor and rounding behaviour are now stated explicitly in Methods with their consequences, and a new table shows that a strict safety certificate (no materially-worse cell on any axis) passes for exactly one dataset at one budget in this study — the audit, not any proxy, is the certificate.

2. I would like more discussion of the precedent for applying feature reduction to lower-dimensional medical datasets, and an explicit statement that the guidance does not extend to the high-dimensional regime.

Response. Both added. Related work now documents the precedent — most directly, the published survival study on the very heart-failure cohort we reuse argued that two of its twelve features suffice, on discrimination-style evidence alone; the Cleveland and Pima datasets were likewise introduced with compact selected-subset models. The Limitations section now carries an explicit, bolded scope statement: the guidance does not extend to the high-dimensional regime ($p \gg 44$; genomic, transcriptomic, radiomic, or wide-EHR feature spaces), where selection is a statistical necessity and retained budgets remain far above the single-digit counts at which we observe the largest harm.

3. I would like at least one worked example showing where reduction would have flipped a real treatment decision for a patient, at a specific clinically relevant threshold.

Response. Added as a new results section, with the cell and threshold fixed *a priori* in a committed analysis configuration before any flip was counted. On the mammographic-mass data at the conventional 10 % biopsy threshold, reducing to the 25 % budget — which on this dataset is a single feature — flips the biopsy decision for 203 of 961 patients (21 %) in the majority of the 30 paired models, including 4 patients with malignant masses moved *from biopsy to no biopsy*; a table of exemplar patients with their feature values and per-threshold decision curves is included, along with sensitivity at $t = 0.05$ and 0.15 . At population scale, the same analysis on the full readmission cohort flips the enrollment decision for 14,324 of 101,763 encounters (14.1 %), including 1,132 true readmissions that lose the intervention. Reliability diagrams show the mechanism (the reduced model’s probabilities compress toward the prevalence), and the per-patient prediction caches behind these counts are committed, so every number regenerates without re-running any experiment.

4. I would like the full Diabetes-130 cohort analyzed, at least for the headline analysis.

Response. Done, beyond the request: the *entire* grid — all rankers, learners, budgets, repetitions, and the recalibration control — now runs on all 101,763 eligible encounters, and every number in the manuscript rests on the full cohort. We also kept the subsample as a robustness arm and report the agreement explicitly (new Section 4.9): point estimates agree in the typical case (median absolute difference 0.004), but 20 of 45 materiality verdicts disagree — in 12 the full cohort resolves an effect the subsample was underpowered to see, in 8 the subsample asserted materiality the full cohort refutes, and in 18 of the 45 the subsample reversed the *sign* of the effect, most consequentially on discrimination (+0.028 full versus −0.028 subsampled). Your concern was substantively correct, and the manuscript now documents what the subsample would have missed.

5. As reproducibility is a central claim of the paper, I ask that the code repository be made available for reviewer inspection.

Response. The repository is public as of this resubmission — not on request, and not deferred to acceptance: github.com/helghareeb/ns-clinical-fs (MIT license). It contains the complete code, the hash-pinned configurations (the originally submitted configuration and aggregates are preserved frozen under `submitted-v1/`), a script that downloads and SHA-256-verifies every raw dataset, the

per-configuration summaries and per-patient caches, the pre-registration with a dated addendum disclosing every post-registration addition, and a cold-start reproduction guide. Every number in the revised manuscript regenerates from committed code, the configuration, and a recorded seed.

Reviewer 2

Thank you for the close read — we applied essentially all of the suggested wording changes, and they made the paper clearer. Specifics worth noting: the overloaded k is resolved by spelling out five-fold cross-validation and reserving k exclusively for the number of retained features (which the revision made load-bearing); the monotonicity claim is now scoped to the mean trajectory and names the panels, acknowledging that the per-dataset ECE curves (left panel) are not monotone; Table 3 is deleted and its content folded into one sentence, as suggested; “a wash on average” and “fairness axes agree” are rewritten in plain terms; “compact” and “wide” are now formally defined at first use ($p \leq 12$ vs. $p \geq 16$) and marked in the meta-analysis figure; “Relation to prior work” is retitled “Relation to established practice”; “co-move” → “move in tandem”; “shuffling” → “permuting”; and the Section-1 and Section-3 line edits are applied as proposed. One deliberate exception: we kept *abstain* (rather than *discard*) because it is the standard term in the selective-prediction literature, and instead added a defining clause at first use (“abstain — decline to issue a prediction”).

Beyond the requested changes

Because the reviews prompted a full rebuild, we took the opportunity to strengthen the paper further: a strict per-dataset safety-certificate table (Section 4.7); leave-one-dataset-out and bootstrap robustness for the cross-dataset correlations; the population-scale flip-rate distribution and its race-group breakdown on the full cohort (decision instability is *not* concentrated on minority groups in this cohort — reported as measured); reliability diagrams for the worked-example cell; a selection-stability analysis whose null result rules out ranking noise as the mechanism of harm; a boxed six-step practitioner checklist; and the enforcement, in code, of the hash-gate the manuscript describes (analysis now refuses to combine results from mismatched configurations). A dated addendum to the deposited pre-registration discloses every post-registration addition explicitly.

A second round of strengthening, and three claims we withdrew

After preparing the response above we continued the audit rather than stopping at the reviewers’ requests, and the additional work changed the paper in ways we want to state plainly — including where it went against us.

The concession Reviewer 1’s line of questioning implied, now measured

The submitted Limitations conceded that AURC is computed from a confidence, $|p - 0.5|$, that is itself a function of the predicted probability, and that establishing selective reliability as *independent* evidence “would require a confidence estimated outside the predicted probability altogether — which we did not run.” We have now run it. For every out-of-fold point we computed the mean Euclidean distance to its ten nearest neighbours in the full feature space, standardised on training-fold statistics only, and used that as the confidence; the model’s predicted probability never enters it.

The result corrects our own headline, and we report it in the Abstract and the Discussion rather than in a limitation. The mean AURC reduction penalty falls from +0.0334 under the probability-scale confidence to +0.0116 under the probability-independent one: **roughly two-thirds of the measured selective-reliability harm was a property of the probability scale.** The remaining third survives, decisively ($p = 7 \times 10^{-103}$ over 2,160 cells) and with no dependence on cohort size, which is what makes it evidence of a genuine loss of error-ordering rather than a restatement of the calibration result. We also tested the obvious model-based alternative first and report that it fails structurally: per-tree disagreement correlates with $|p - 0.5|$ at $\rho = -0.991$, because a forest disagrees most where its mean prediction sits near the decision boundary.

“You did not try a flexible enough recalibrator”

The submitted version rested the claim that recalibration cannot recover selective reliability on isotonic regression alone. We added temperature scaling and beta calibration under the same leakage-safe inner-cross-validation protocol and ordered all five recalibrators by flexibility on one matched grid. Calibration-error recovery rises *strictly monotonically* with flexibility, from +0.0054 at one parameter to +0.0193 for the nonparametric recalibrator (Spearman +1.00), reaching 27.8 % of the uncalibrated baseline; selective-reliability recovery does not follow, and its correlation with flexibility runs the opposite way (−0.90), with the largest movement worth 3.1 % of baseline. The failure to recover is therefore not a limitation of the recalibrator. Net-benefit recovery does rise monotonically, and we report that rather than describe it as flat — its magnitude, 0.9 % of baseline, is what makes it negligible, not its direction.

A causal test of the mechanism, whose pre-registered prediction failed

Both reviewers’ concerns turn on *when* reduction is safe, so we tested our proposed explanation causally instead of correlationally, and pre-registered the prediction before computing. If diffuse signal is what makes reduction harmful, then padding a dataset with pure-noise features — holding rows, true features, folds and the *absolute* budget fixed — should increase the harm.

It did not. Dilution lowered the penalty (cleveland, +0.054 $\rightarrow$ +0.006 at ten-fold padding), reversed its sign (SPECTF, +0.004 $\rightarrow$ −0.037), or left it unmoved (mammographic, +0.086 $\rightarrow$ +0.081). **Reduction defends against noise padding**, because the ranker keeps recovering the true features while the full model degrades. We report the refutation as measured. It also sharpens the account: what distinguishes a harmed dataset is not how much noise it contains but how many features carry its signal, an effective dimensionality that no single concentration index captures across widths differing by two orders of magnitude.

Four further controls, two of which returned nothing

Since the response above was drafted we added four more analyses, and we summarise them here because two are negative and a reader is entitled to see those alongside the positive ones.

The reliability axes are not one scale, and now we can say by how much. Orienting all five axes so that positive means worse and correlating the reduction penalties across the 108 cells, the axes fall into two groups: those defined on the *ordering* of predicted probabilities move together (AURC against conformal set size, +0.89), those defined on the probability *scale* move together (ECE against selective-ECE, +0.97), and the two groups are nearly independent (ECE

against AURC, +0.30). This is the same division the recalibration ladder implies mechanically — recalibration repairs a scale and cannot repair an ordering — arrived at from a different direction.

The retained-count association is partly a collinearity, and we say so. Because the fractional rule sets $k = \lceil 0.25p \rceil$, a small retained count and a narrow dataset are hard to separate. Holding the budget fixed at $k = 8$ across the clinical datasets for which that is genuinely a reduction, harm still rises with width ($\rho = +0.77$, $p = 0.016$). What a fixed budget discards therefore matters independently of what it keeps. This weakens the strong form of a claim we made ourselves, and the manuscript now states the weaker form.

Two negatives. We asked whether the harm that survives a probability-independent confidence sits preferentially in the datasets with concentrated signal or small retained counts: it does not (+0.21, -0.16, -0.13, all null), so these are two findings rather than one mechanism seen twice. And we made a deliberate attempt to refute our own claim that no measured proxy certifies a budget in advance, using the ranker’s concentration curve, with both thresholds locked in a deposited pre-registration before computing. It fails: sensitivity 1.00 but specificity 0.15, flagging 33 safe budgets as unsafe across 80 cells. A screen that says “unsafe” almost everywhere is not a certificate, and the negative claim stands.

Three claims withdrawn, and why we are telling you

A pre-submission audit of the manuscript against its own released results found three statements we could not defend, and we removed them rather than qualify them.

1. An exploratory claim that the four high-dimensional datasets are *ordered* from least to most harmed by signal concentration. Tested across three rankers and three concentration statistics it held in two of nine cases, and in *none* of twenty-one once we corrected the statistic for the width incomparability that caused the failure. A concentration column in our own released results refuted it. Only the prostate-versus-arcene contrast, which is robust across every ranker, budget and statistic, is retained.
2. A sentence in the Discussion recommending that budget decisions be “guided by the feature-count / intrinsic-dimension screen above.” That screen is precisely what this revision withdrew in response to Reviewer 1; the sentence was stale text pointing readers at a criterion the same manuscript retracts two sections earlier.
3. The description of net-benefit recovery as flat, corrected above.

We mention these because the alternative — leaving them in, since no reviewer had asked — is the failure mode this protocol exists to prevent. Every number in the manuscript is regenerated from a committed configuration and a recorded seed, every result file’s calibration hash resolves to a configuration shipped in the repository, and every column in every released results file is either cited in the paper or declared out of scope with a stated reason.

We believe the revised manuscript is substantially stronger than the original, in large part because of these reviews, and we hope it now meets the bar for acceptance.